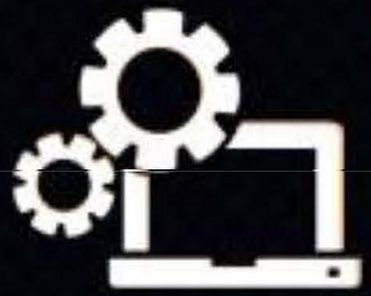
CS & IT ENGINEERING

DIGITAL LOGIC

MINIMIZATION



Lecture No. 3



By - CHANDAN SIR

Recap of Previous Lecture



Basic of Boolean

Questions

Topics to be Covered



K-Map

- K-Map
- Questions Practice
- Discussion

Duality

$f = f^D$

Self Dual

$2^{2^{n-1}}$

→ Self Dual Expression

$$f = AB + BC + AC$$

$$f^D = (A+B)\cdot(B+C)\cdot(A+C)$$

$$= (A+B)\cdot(A+C)\cdot(B+C)$$

$$= (A+BC)\,(B+C)$$

$$= AB + AC + BC\cdot B + BC\cdot C$$

$$f^D = AB + AC + BC$$

$$f^D = f$$

Self Dual



Topic : K-Map

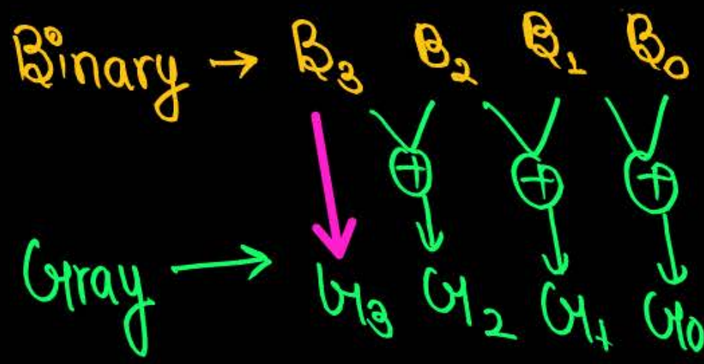
↳ By this we can minimize the function.



Topic : K-Map Basics

Minimization by K-Map

- Based on gray code
- Gray code ⟶ It is the code in which successive numbers are differ by one bit



Examplee→ Binary → 1 1 0 0 1 0 1

Gray → 1 0 1 0 1 1 1

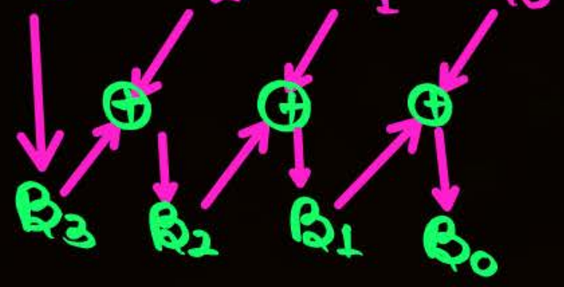
Q

Binary → 110010110

Gray → 101011101

Gray code → G_3 G_2 G_1 G_0

Binary code → B_3 B_2 B_1 B_0



Q Gray code ⇒ 1011100

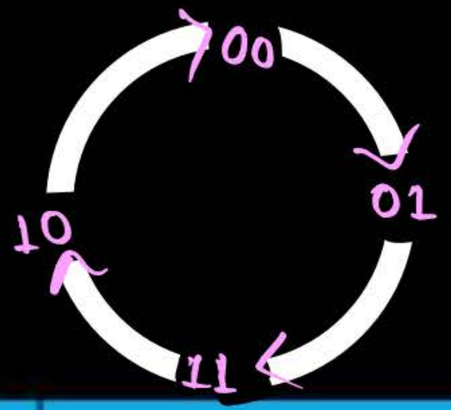
Binary ⇒ 1101000

Q Gray code → 1 1 0 0 1 0 1

Binary → 1 0 0 0 1 1 0



Topic : K-Map Basics



Gray Code

1> unity haming Distance code

2> Cyclic code

3> Reflecting code

Decimal	Binary	Gray Code
0	0 0	0 0
1	0 1	0 1
2	1 0	1 1
3	1 1	1 0

B ≡ B



Topic : K-Map Basics

Gray Code

Decimal	Binary	Gray Code
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100



Topic : K-Map Basics

$f(A, B)$

A \ B	0 ($\bar{B}$)	1 (B)
0 ($\bar{A}$)	$\bar{A}\bar{B}$ 00 0	$\bar{A}B$ 01 1
1 (A)	$A\bar{B}$ 10 2	AB 11 3



Topic : K-Map Basics

$f(A, B, C)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	000 0	001 1	011 3	010 2
A 1	100 4	101 5	111 7	110 6



Topic : K-Map Basics

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0000 0	0001 1	0011 3	0010 2
$\bar{A}B$ 01	0100 4	0101 5	0111 7	0110 6
AB 11	1100 12	1101 13	1111 15	1110 14
$A\bar{B}$ 10	1000 8	1001 9	1011 11	1010 10

$f(A, B, C, D)$



Topic : K-Map Basics

16 group = 2^4 → 4 Variable Reduce

8 group = 2^3 → 3 Variable Reduce

4 group = 2^2 → 2 Variable Reduce

2 group = 2^1 → 1 Variable Reduce

1 group = 2^0 → '0' Variable Reduce



Topic : K-Map Basics

Rule of Minimization

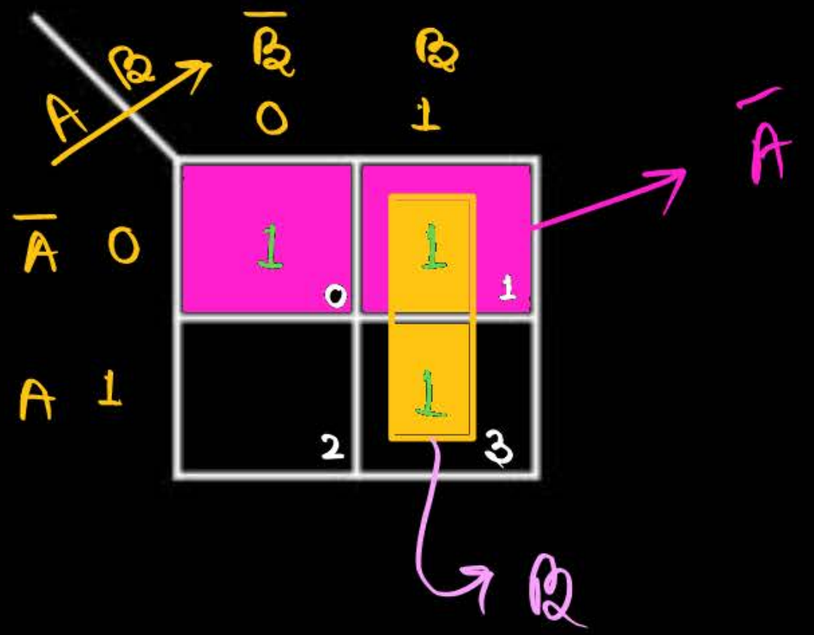
CI Baba Rule

↳ Kam se kam group banana hai and Bade se bada group banana hai.

Terms will minimize

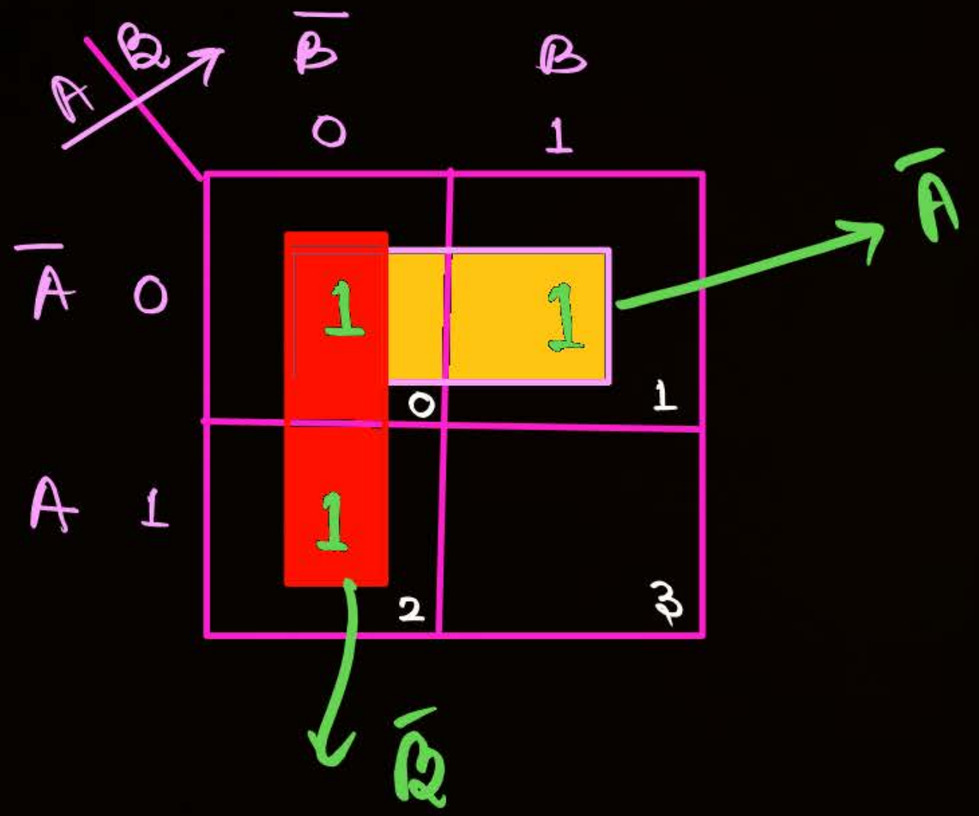
Variable will minimize

#Q. $f(A, B) = \bar{A}\,\bar{B} + \bar{A}B + AB = \sum m\,(0, 1, 3)$



$\bar{A} + B$

Q $f(A,B) = \Sigma m(0,1,2)$



$\bar{A}+\bar{B}$

Ans

$\bar{A}\bar{B}+\bar{A}B+A\bar{B}$

$\bar{A}(\bar{B}+B)+A\bar{B}$

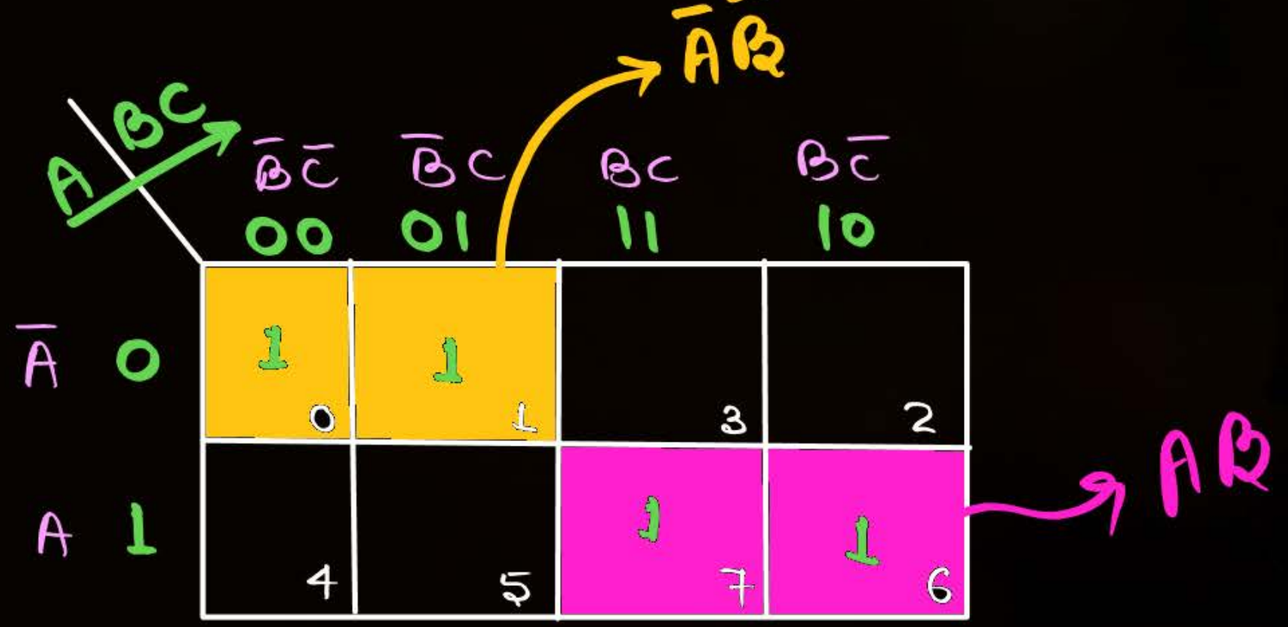
$\bar{A}+A\bar{B}$

$(\bar{A}+A)(\bar{A}+\bar{B})$

$\bar{A}+\bar{B}$

Ans

Q $f(A,B,C) = \Sigma m(0,1,6,7)$



$\bar{A}\bar{B} + AB$

$\bar{A}\bar{B}$

AB

Q $f(A,B,C) = \Sigma m(0,1,3,6,7)$

HW

Q $f(A,B,C) = \Sigma m(0,1,3,5,6,7)$



2 Minute Summary

K-Map

Questions Practice

Discussion

Thank you

GW

Soldiers !

